

Instructor: Dr. Jim D. Pham
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Office Hrs.: By Appointment
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Section: CS/CNET-147-0 (086894)

Course Description:

This course introduces a variety of the tools and concepts used while working with a UNIX/Linux-based computer system. It introduces UNIX shells, comparing, and contrasting the major ones. Students will learn to write shell scripts using basic commands, regular expressions, vi, sed, and awk. They will then use those tools to write scripts for various shells such as the Bourne, C, Korn, and Bash shells. Other scripting tools such as Perl and Python will also be explored. Students will write shell script programs to exercise their understanding of tools and concepts. This course will be taught using a combination of lectures, demonstrations, discussions, and hands-on labs.

Prerequisites: NA**Recommended textbook:**

- Shell Scripting Tutorial [The Shell Scripting Tutorial](#) (Free)
- Shell Scripting Tutorial [Shell Scripting Tutorial - Tutorialspoint](#) (Free)
- Practical Guide to Linux Commands, Editors, and Shell Programming, A, 4th Edition by Mark G. Sobell, Matthew Helmke ISBN-13: 978-0-13-477460-2

Student Learning Outcomes

- After completing the course, students will be able to:
- Construct regular expressions containing metacharacters for use at the command line or in shell scripts.
- Demonstrate proficiency using grep to search for regular expressions in files.
- Demonstrate proficiency using the vi editor for creating and editing text files.
- Demonstrate proficiency using sed to perform editing tasks useful in shell scripts.
- Demonstrate proficiency using awk for manipulating data and generating reports.
- Write shell scripts using C-shell, Bourne shell, Korn Shell, and bash, demonstrating knowledge of the appropriate environment commands to accomplish tasks such as data search and retrieval, file editing, system administration, and backup.

Teaching Methods:

1. Lectures: The presentation slides and assignments will be accessed through Ohlone's Canvas Course Management System. See Instructional Resources below.
2. Assignments: End-of-chapter activities and online activities will be assigned weekly to reinforce the material in the text. These assignments may require the application of various software packages.
3. Quizzes: Weekly unannounced quizzes may be given to help ensure students keep up with assigned material.
4. Exams: Two exams will be given: a) Midterm; and b) Final Exam. The exams will be delivered online, they will be timed, and will test assigned readings and material discussed in class. The final exam will be comprehensive. However, the instructor reserves the right to retest on material that was not appropriately comprehended.
5. Participation: Student participation will be graded by the level of class participation on the Discussion Board.

Email:

If you send me an email, please put **course number – name – topic** into the email **Subject** field.
For example, CS/CNET 147 – Name – Issue

Labs and Assignments:

Lab programs are required for this course. Please be ready to discuss the techniques demonstrated in the online lecture and explained in the textbook via Canvas.

Exams and Tests:

There will be Quizzes, Midterm, and a comprehensive Final Exam. For all exams, you will be allowed to use books and notes. The questions will be multiple-choice, true/false, short answer fill-ins, and programming. All exams will be timed.

Policy on Student Work:

Labs and assignments are to be done individually unless a group project is specified. You are encouraged to talk to fellow students about concepts and methods, but you may not copy someone else's work. I adhere to the **Ohlone College Policy on Academic Integrity**.

Grading:

Labs:	30%
Participation:	10%
Chapter Quizzes:	10%
Midterm Exam:	20%
Final Exam:	30%

Grading Scale:

A = 90-100
B = 80-89
C = 65-79
D = 55-64
F = <55

Course Schedule: The following is a recommended schedule. The instructor reserves the right to make any schedule changes deemed necessary.

Session	Date	Activity
1	06/13/2022 06/15/2022	- Using the Shell, Getting Help, Text Utilities, - Configuring the Shell
2	06/20/2022 06/22/2022	- A First Script - Variables
3	06/27/2022 06/29/2022	- Wildcards - Escape Characters
4	07/04/2022 07/06/2022	- Holiday – Independence Day - Loop
5	07/11/2022 07/13/2022	Midterm - Test and Case
6	07/18/2022 07/20/2022	More Variables
7	07/25/2022 07/27/2022	- External Programs - Functions
8	08/01/2022 08/03/2022	- Brief on Perl, AWK, Sed Editor Final Exam